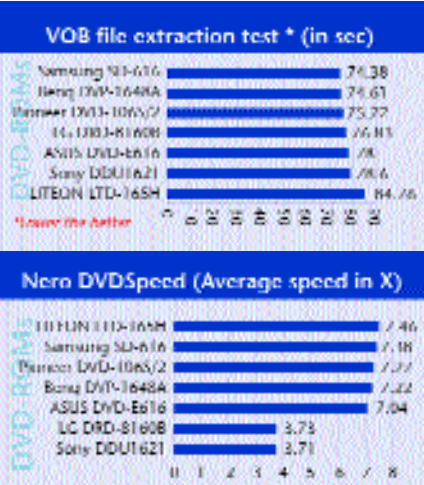




software CDs or CDs such as the Digit Interactive CDs, which contain a lot data in varying sizes.

A CD containing assorted data in the form of files ranging from 1 KB to 100 MB was copied on the hard disk and the time was noted. The LITEON LTD-165H was the fastest here, taking 144.14 seconds to copy the data. The ASUS DVD-E616 and Samsung SD-616 were just seconds behind, logging 146.46 and 147.76 seconds, respectively. The LG DRD-8160B was the slowest DVD drive and took 210.12 seconds to copy the CD to the hard disk.

Disk Registration: The disk registration test shows how fast the drive can identify a CD or DVD. This test will be especially interesting to those who share their drives on the network for others to use.

We used the Digit Mindware CD (October issue) for this test. The Samsung SD-616 was the slowest, taking 15.13 seconds, whereas the LG-DRD 8160-B topped the test by taking only 7.60 seconds to recognize the CD. The Benq DVP-1648A finished second by identifying the CD in just 8.89 seconds.

Exact Audio Copy: We extracted three songs from an Audio CD, such that the first, the middle and the last tracks of an 80-minute CD were accessed. Since access speeds vary as we traverse the radius of a CD, this test would give us a good indication of the overall performance of the drive. Exact Audio Copy was used to evaluate the same. The overall winner of this test was the LITEON LTD-165H, which took just 33.91 seconds to rip all the three tracks. The slowest however, was the Samsung drive, which performed quite poorly with a score of 83.68 seconds.



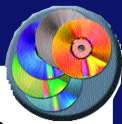
Read tests (CDs): Though all the DVD-ROM drives are rated between 40 and 48x, we observed noticeable differences in performance while reading CDs.

Assorted Read: This tests the DVD-ROM drive's ability to read an assortment of files that come in varying sizes. This test will interest those who often use

SiSoft Sandra 2002 Pro: The three tests carried out using SiSoft Sandra 2002 Pro were sequential read, random read and average access time. This benchmark gives an overall view of how a drive will perform under different conditions.

The sequential read speed is the continuous read

Hybrid DVDs



Nine leading consumer electronics manufacturers have jointly established the basic specifications for a next-generation large-capacity optical disc video recording format called Blue-ray Disc. These DVDs can hold six times the data as current standard discs. The new format will store more than 13 hours of film, compared with the current limit of 133 minutes.

High Density-DVD no doubt, is the enhanced form of current DVD format. Blue lasers have a much shorter wavelength than the red lasers used in current DVD players, allowing for much greater disc capacity. The new high definition format allows a capacity of 4.7 GB to 15 GB on a single-sided, single-layered disc and a capacity of 8.5 GB to 30 GB on a single-sided, double-layered disc. Some feel be more definite! that this provides a cost-effective upgrade path for media vendors and a backward-compatible solution for all those who've already built DVD libraries.

Current DVD players use 650 nanometre (nm) red lasers, while CD players use 780 nm diodes. It will not be possible to play Blue-ray discs with existing DVD and CD players, since these don't contain 405 nm lasers. Blue-ray drives will need to contain 780, 650 and 405 nm lasers in order to be compatible with CD and DVD discs.

speed ability of the drive and is useful while reading large single files such as ISO images.

The winners in this category were the Benq DVP-1648, Pioneer DVD-106S/2 and LITEON LTD 165-H with a read speed of 9.216 MBps each. These drives are thus ideally suited for those who copy a lot of large single files from optical media to the hard drive or are into image editing. The LG-DRD 8160 B performed poorly with a sequential read speed of 4.096 MBps—a slow solution when compared to its competition.

The random speed determines the ability of a DVD-ROM drive to randomly read data from an optical media. This test will interest those who have a lot of clipart to be copied to the hard drive from either CD or DVD media, such as DTP shops. The two drives in competition with each other were the Pioneer DVD-106S/2 and the Benq DVP-1648A with a random read time of 1119 and 1100 KBps, respectively. This was the only test in which the Pioneer drive—which had won past DVD-ROM comparisons—shone. The poor performer in this test was the Sony DDU 1621-C0, logging 513 KBps only.

The access time tells us how fast the drive can access a particular track and start reading the information. Here, the Pioneer DVD-106S/2 won by taking just 50 ms. The Benq DVP-1648A was a close second with 51 ms—a 1 ms difference that makes little practical difference. Sony was the slow respondent again and probably was the only drive to cross the 100 ms mark by logging 111 ms as access time.

What it all leads to

The competition between the CD-Writers was really tough. We got some unexpected results, where drives with a faster rating were overtaken by drives with a slower rating and older technology. What this tells us is that 32x is still where it's at—it not only performs optimally, but also costs much lesser.